Light

GCSE Computer Science — Memory and Storage (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What happens to the data stored in RAM when the computer is switched off?

- **A.** The data is lost
- **B.** The data is saved automatically to the hard drive
- **C.** The data remains stored permanently
- **D.** The data is compressed to save space

Q2 (1 mark)

What type of data is typically stored in ROM?

- **A.** User documents and photos

- ○ **B.** The BIOS/boot-up instructions needed to start the computer
- ○ **C.** Temporary data from running programs
- ○ **D.** Downloaded files from the internet

Q3 (1 mark)

What is virtual memory?

- ○ **A.** A type of cloud storage accessed over the internet
- ○ **B.** Extra RAM chips that can be added to a computer
- ○ **C.** A section of secondary storage used as an extension of RAM when RAM is full
- ○ **D.** Memory used only by virtual reality applications

Q4 (1 mark)

Which type of secondary storage uses spinning platters and read/write heads?

- ○ **A.** Magnetic Hard Disk Drive (HDD)
- ○ **B.** Solid State Drive (SSD)
- ○ **C.** USB flash drive
- ○ **D.** Optical disc (CD/DVD)

Q5 (1 mark)

What is a key advantage of Solid State Drives (SSDs) over Hard Disk Drives (HDDs)?

- ○ **A.** SSDs have larger storage capacity
- ○ **B.** SSDs have faster access speeds and no moving parts
- ○ **C.** SSDs are less expensive per gigabyte
- ○ **D.** SSDs use magnetic storage technology

Q6 (1 mark)

How does an optical disc (such as a CD or DVD) store data?

- ○ **A.** Using magnetic fields on a spinning platter
- ○ **B.** Using flash memory chips
- ○ **C.** Using a laser to create pits and lands on a reflective surface
- ○ **D.** Using radio waves transmitted to the disc

Q7 (1 mark)

How many bits are in one byte?

- ○ **A.** 2 bits
- ○ **B.** 4 bits
- ○ **C.** 16 bits
- ○ **D.** 8 bits

Q8 (1 mark)

How many bytes are in 1 kilobyte (KB)?

- ○ **A.** 1000 bytes
- ○ **B.** 1024 bytes
- ○ **C.** 100 bytes
- ○ **D.** 10000 bytes

Q9 (1 mark)

What is the main purpose of RAM in a computer system?

- ○ **A.** To store the operating system permanently
- ○ **B.** To hold currently running programs and data being actively used
- ○ **C.** To provide long-term backup storage
- ○ **D.** To display images on the monitor

Q10 (1 mark)

Which secondary storage device is most suitable for transferring files between different computers?

- ○ **A.** Internal Hard Disk Drive
- ○ **B.** RAM module
- ○ **C.** USB flash drive

- ○ **D.** CPU cache

Q11 (1 mark)

Which type of secondary storage typically offers the largest capacity at the lowest cost per gigabyte?

- ○ **A.** Magnetic Hard Disk Drive (HDD)
- ○ **B.** USB flash drive
- ○ **C.** Solid State Drive (SSD)
- ○ **D.** Optical disc (DVD)

Q12 (1 mark)

Which statement about ROM is correct?

- ○ **A.** ROM can be easily written to by the user
- ○ **B.** ROM is non-volatile and retains data without power
- ○ **C.** ROM loses its data when power is removed
- ○ **D.** ROM is used to store temporary program data

Q13 (2 marks)

A computer with 4GB of RAM is running several large applications that require 6GB of memory in total.
Explain what happens and how it affects performance.

- ○ **A.** The computer crashes immediately because there isn't enough RAM
- ○ **B.** The extra applications are automatically deleted to fit in RAM
- ○ **C.** The applications automatically compress themselves to fit in the available RAM
- ○ **D.** Virtual memory is used, moving less-used data to secondary storage, but this slows down the system because secondary storage is slower than RAM

Q14 (2 marks)

A file has a size of 2,500,000 bytes.
What is this size in megabytes (MB)?

- ○ **A.** 0.25 MB
- ○ **B.** 25 MB
- ○ **C.** 2.5 MB
- ○ **D.** 250 MB

Q15 (2 marks)

A photographer needs to store photos while travelling to remote locations with rough conditions.
Which storage characteristic should be the highest priority, and which device best meets this need?

- ○ **A.** Durability — an SSD or flash drive has no moving parts and is more resistant to physical shock
- ○ **B.** Capacity — a large HDD would store the most photos

- ○ **C.** Cost — optical discs are cheapest per gigabyte
- ○ **D.** Speed — the fastest storage will prevent photo corruption

Q16 (2 marks)

A bitmap image is 100 pixels wide and 50 pixels high, with a colour depth of 24 bits.
What is the file size of this image in bytes (ignoring metadata)?

- ○ **A.** 5,000 bytes
- ○ **B.** 15,000 bytes
- ○ **C.** 120,000 bytes
- ○ **D.** 150,000 bytes

Q17 (2 marks)

A sound file is recorded at a sample rate of 44,100 Hz.
What does this sample rate mean?

- ○ **A.** The sound plays for 44,100 seconds
- ○ **B.** The file size is 44,100 bytes
- ○ **C.** The sound has 44,100 different volume levels
- ○ **D.** 44,100 samples of the sound wave are taken every second

Q18 (2 marks)

The ASCII character set uses 7 bits to represent each character.
How many different characters can 7-bit ASCII represent?

- ○ **A.** 7 characters
- ○ **B.** 49 characters
- ○ **C.** 128 characters
- ○ **D.** 256 characters

Q19 (2 marks)

A business needs to store important financial records that must be kept for 7 years.
Which storage consideration is most important for this use case?

- ○ **A.** Reliability and longevity — the data must remain intact and readable for many years
- ○ **B.** Access speed — the records must be retrieved instantly
- ○ **C.** Portability — the records must be carried between offices
- ○ **D.** Cost per gigabyte — the cheapest option should always be chosen

Q20 (2 marks)

An SSD can access data in approximately 0.1 milliseconds, while an HDD takes approximately 10 milliseconds.
Explain why this difference exists.

- ○ **A.** SSDs use larger storage platters that spin faster
- ○ **B.** HDDs have faster electronic circuits than SSDs
- ○ **C.** HDDs compress data more, requiring extra time to decompress

- ○ **D.** SSDs access data electronically with no moving parts, while HDDs must physically move read/write heads to the correct position

Q21 (2 marks)

A music streaming service offers 50 GB of storage for offline downloads. How many 5 MB songs can be stored (assuming no overhead)?

- ○ **A.** 100 songs
- ○ **B.** 10,000 songs
- ○ **C.** 1,000 songs
- ○ **D.** 100,000 songs

Q22 (2 marks)

A software company needs to distribute a 20 GB video game. Which optical disc format would be most suitable?

- ○ **A.** CD (700 MB capacity)
- ○ **B.** Single-layer DVD (4.7 GB capacity)
- ○ **C.** Blu-ray disc (25 GB single-layer capacity)
- ○ **D.** Dual-layer DVD (8.5 GB capacity)

Q23 (2 marks)

A user upgrades their computer's RAM from 4 GB to 16 GB. What improvement are they most likely to notice?

- ○ **A.** Files will be saved more quickly to the hard drive
- ○ **B.** The internet connection will be faster
- ○ **C.** Games will have better graphics quality
- ○ **D.** The computer can run more applications simultaneously without slowing down

Q24 (3 marks)

A stereo audio file is recorded for 3 minutes at a sample rate of 44,100 Hz with a bit depth of 16 bits.
Calculate the file size in megabytes (MB).

- ○ **A.** Approximately 31.8 MB
- ○ **B.** Approximately 15.9 MB
- ○ **C.** Approximately 63.5 MB
- ○ **D.** Approximately 127 MB

Q25 (3 marks)

A graphic designer must reduce the file size of images for a website while maintaining acceptable quality.
Analyse the effects of reducing resolution and reducing colour depth.

- ○ **A.** Reducing resolution has no effect on file size; only colour depth matters
- ○ **B.** Reducing colour depth has no effect on quality; only resolution matters
- ○ **C.** Reducing resolution decreases the number of pixels (affecting detail/sharpness), while reducing colour depth limits the range of

colours (potentially causing banding); both reduce file size but affect different aspects of quality

○ **D.** Both changes increase file size but improve quality

Q26 (3 marks)

A video production company needs to choose storage for three different purposes:

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• Editing current projects (frequent read/write operations)
• Archiving completed projects (long-term storage, rarely accesse
• Transporting project files to client locations
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Analyse which storage types best suit each purpose.

○ **A.** Use the same HDD for all three purposes because it's cheapest

○ **B.** Editing: SSD (fast access speeds for real-time editing); Archive: HDD or tape (high capacity, low cost per GB); Transport: USB SSD or flash drive (portable, durable)

○ **C.** Use cloud storage exclusively for all purposes regardless of internet speed

○ **D.** Editing: Optical disc; Archive: USB flash drive; Transport: Internal HDD

Q27 (3 marks)

A computer's hard drive LED is constantly active and the system has become extremely slow, even though no files are being opened or saved by the user.
Analyse what is likely happening and explain the cause.

○ **A.** The hard drive is failing and needs to be replaced immediately

○ **B.** A virus is deleting all files from the hard drive

○ **C.** The computer is automatically uploading all files to the cloud

○ **D.** The system has insufficient RAM, causing excessive virtual memory usage where data is constantly swapped between RAM and the hard drive — a condition called disk thrashing

Q28 (3 marks)

Explain why all types of data (text, images, sound, video) are ultimately stored as binary in computer systems.

○ **A.** Computer hardware uses electronic circuits that have two stable states (on/off, high/low voltage), which naturally represent binary digits (0 and 1); all data must be converted to binary to be stored and processed by these circuits

○ **B.** Binary is simply an arbitrary choice that could easily be replaced with decimal

○ **C.** Binary takes up less storage space than any other number system

○ **D.** Binary was chosen because it's easier for humans to understand than other systems

Q29 (3 marks)

A sound recording increases its bit depth from 8-bit to 16-bit. Analyse how this change affects the audio quality and file size.

○ **A.** The file size doubles but quality remains the same

○ **B.** The sample rate doubles, capturing more samples per second

○ **C.** The number of possible amplitude levels increases from 256 to 65,536, providing much more precise representation of sound wave

amplitudes with less quantisation noise; file size doubles as each sample uses twice the bits

- ○ **D.** The file size halves because 16-bit is more efficient

Q30 (3 marks)

A smartphone has 128 GB of storage. After installing the operating system (12 GB), essential apps (8 GB), and taking 400 photos averaging 4 MB each, how much storage remains available for additional content?

- ○ **A.** Approximately 96.4 GB
- ○ **B.** Approximately 108 GB
- ○ **C.** Approximately 120 GB
- ○ **D.** Approximately 106.4 GB

Total: **55** marks • Time: **30 min**

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